Ayan Ahmed

Software Engineer, Google

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Education

2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78

2015 **XII - Karnataka Board,** *KLE Society's Independent PU College, Bangalore* 97.30 %

2013 **X - ICSE,** *B P Indian Public School, Bangalore* 96.33%

Professional Experience

Dec 2022 - Present **Software Engineer, Level IV, Google LLC, New York**
- Working on Keep, a notetaking editor in Google Workspace.

Aug 2020 - Nov 2022 **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
- Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
- Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
- Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.

May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.

May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis,* ***Language Technologies Institute, Carnegie Mellon University***
Machinated a novel technique for paraphrase generation using the von Mises-Fisher (vMF) Loss on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by *Prof. Yulia Tsvetkov*.

May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
Research Intern, ***Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore***
Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by Prof. Veni Madhavan.

Publications and Patents

[Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan*, Sachin Kumar, John Wieting, Yulia Tsvetkov*
In **Workshop on Multilingual Representation Learning**,
The 2021 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2021**)

[Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
In **ACM CODS-COMAD 2020** - Young Researchers' Symposium

[Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, ***Monisha Jegadeesan****, Amrit Singhal, Kush Kumar Singh, Kundan Krishna*
In ACM User Interface Software and Technology Symposium 2019 (**ACM UIST 2019**)

[Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, ***Monisha Jegadeesan****, Amrit Singhal, Kush Kumar Singh, Kundan Krishna*
Filed at the US PTO (Application Number: 16/247,235)

[Publication and Poster] **Leveraging Ontological Knowledge for Neural Language Models (Paper, Poster)** *Ameet Deshpande, **Monisha Jegadeesan***
In **ACM CODS-COMAD 2019** - Young Researchers' Symposium

Projects

Real-Time AI-Powered Customer Support Chatbot (2024)

Designed and implemented a scalable AI-powered chatbot leveraging large language models and retrieval-augmented generation (RAG) to instantly respond to customer queries. Developed an end-to-end ML pipeline including data ingestion, document embedding using OpenAI embeddings, and vector database creation with ChromaDB. Integrated the chatbot into production using LangChain and OpenAI API, reducing human agent workload by 35% and improving response latency to near real-time, demonstrating proficiency in AI system integration, model deployment, and optimization.

Tech: LangChain, OpenAI API, ChromaDB, Python, RAG, LLM integration

Customer Sentiment Classification Using NLP (2024)

Built a machine learning pipeline to analyze and classify customer sentiment from unstructured social media reviews to support business decision-making. Executed comprehensive data preprocessing including text cleaning, tokenization, and stopword removal, followed by feature engineering with TF-IDF vectorization. Designed and trained a logistic regression model optimizing classification performance to 89% accuracy. This project showcased capabilities in supervised learning, NLP, model evaluation, and preparing data-driven insights aligned to product improvement strategies.

Tech: Python, NLTK, TF-IDF, Logistic Regression

Predictive Maintenance Model for Industrial Equipment (2024)

Developed a predictive maintenance solution leveraging machine learning to forecast machinery failures using time-series sensor data from industrial IoT devices. Implemented scalable data pipelines for sensor data ingestion and preprocessing. Engineered features from temperature, vibration, and pressure metrics, and trained a Random Forest classifier. Evaluated and optimized the model using F1-score and precision metrics. The deployed model reduced unexpected equipment downtime by 28%, emphasizing expertise in ML development, data engineering, and production model deployment.

Tech: Python, Random Forest, Time-Series Analysis, IoT Sensor Data

Employee Attrition Prediction and Dashboard for HR Analytics (2024)

Designed an end-to-end data-driven solution to identify and predict drivers of employee attrition. Processed structured HR datasets using SQL and Python for data cleaning and exploratory analysis. Built and validated a logistic regression model to estimate attrition probabilities, enabling targeted retention strategies. Developed an interactive Power BI dashboard for business stakeholders to monitor attrition trends in real-time. The solution contributed to a 12% reduction in attrition by informing policy changes, demonstrating strengths in data engineering, predictive modeling, cross-team collaboration, and stakeholder communication.

Tech: SQL, Python, Logistic Regression, Power BI

About :

I am an AI Engineer specializing in designing, developing, and deploying scalable machine learning models and AI/ML solutions aligned with business goals. Proficient in Python and ML frameworks such as TensorFlow, PyTorch, Scikit-learn, and Hugging Face, I build end-to-end ML pipelines including data ingestion, preprocessing, training, testing, and deployment. Experienced in integrating AI/ML models into production systems using APIs, microservices, and cloud platforms (AWS, Azure, GCP) with a strong focus on MLOps practices including CI/CD, Docker, and Kubernetes. Skilled in NLP, LLMs, and generative AI, I optimize models for performance, latency, and scalability while collaborating with cross-functional teams to deliver data-driven solutions. My approach emphasizes clean and scalable data pipelines, model evaluation using statistical techniques, and continuous research on emerging AI/ML trends to drive enterprise-scale innovation.